Geometry
Unit 2
Lesson 5 Homework

Name Answer Key____________
Date____________________
Period___________________

Complete each table for the conditional statement.

If an angle is classified as a right angle, then it has a measure of $90°$.			
	Relationship to Conditional	**Statement**	**True or False**
1.	Converse	If an angle has a measure of $90°$, then it is a right angle.	True
2.	Inverse	If an angle is not classified as a right angle, then it does not measure $90°$.	True
3.	Contrapositive	If an angle does not measure $90°$, then it is not a right angle.	True
Write the conditional using "if and only if".			
4.	An angle is classified as a right angle if and only if it measures $90°$.		

If two angles have the same measure, then they are congruent			
	Relationship to Conditional	**Statement**	**True or False**
5.	Converse	If two angles are congruent, then the angles have the same measure.	True
6.	Inverse	If two angles do not have the same measure, then they are not congruent.	True

7.	Contrapositive	If two angles are not congruent, then the angles do not have the same measure.	True
Write the conditional using "if and only if".			
8.	Two angles have the same measure if and only if the angles are congruent.		

9. Write a conditional statement that is biconditional.

Answers may vary. A possible correct answer: If the sum of the interior angles of the polygon is $180°$, then the polygon is a triangle.

10. Write a conditional statement that is **<u>not</u>** biconditional.

Answers may vary. A possible correct answer: If it is raining, then there are clouds in the sky.